

A software company has been contracted to develop an **Online Hospital Management System**. The project manager has allocated the following resources:

- Number of software developers = **8**
- Working days = **22 days**
- Working hours per day = **8 hours**
- Total project budget = **\$180,000**
- Estimated effort required to complete the project = **1,200 hours**

Based on historical project data, the company has determined the following productive utilization rates at different CMM maturity levels:

CMM Level	Productive Utilization
Level 1 – Initial	58%
Level 2 – Managed	68%
Level 3 – Defined	78%
Level 4 – Quantitatively Managed	87%
Level 5 – Optimizing	93%

Assume that the **project budget is fully spent** at each maturity level.

- Calculate the **gross project capacity (in hours)**.
- Calculate the **effective project capability (productive hours)** for each CMM maturity level.
- Calculate the **Project Capability Percentage** for each CMM level.
- Calculate the **productive budget utilization** at each CMM level.

Based on your calculations, explain how increasing CMM maturity improves: project capability, budget utilization, project predictability, and software quality.